Hadronic Physics I

Geant4–11.1 reference - Based on previous Geant4 courses

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Covered topics

- Hadronic physics: final state models *vs.* cross sections
- Hadronic physics framework
- Capture at rest
- Neutron physics
- Hadronic physics in reference physics lists

… to be followed by Hadronic Physics II at this course

Geant4 particles and hadrons (1/3)

In `Geant4` several `G4ParticleDefinitions` exist, i.e. "stable" particles described by their properties (mass, charge, …)

- Leptons: $e^{\pm}$ (`G4Electron`, `G4Positron`), $\mu^{\pm}$ (`G4MuonMinus`, `G4MuonPlus`), …
- Bosons: γ (`G4Gamma`), `G4OpticalPhoton`
- `G4Geantino`, i.e. a particle with no processes except transportation
- "Stable" hadrons: $\pi^{\pm}$ (`G4PionPlus`, `G4PionMinus`), p (`G4Proton`), n (`G4Neutron`), …
- Light ions: deuterium (`G4Deuteron`), tritium (`G4Triton`), …
- Hyper and anti-hyper nuclei, i.e. nuclei with at least one hyperon (baryons with strange quark content): `G4HyperTritium`, `G4HyperAlpha`, …
- Other ions: `G4GenericIon` describes all other ions

Geant4 particles and hadrons (2/3)

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Bash - Example: inspect π^{+} properties

```
/particle/select pi+     # UI Command
/particle/property/dump  # UI Command

--- G4ParticleDefinition ---
Particle Name : pi+
PDG particle code : 211
Mass [GeV/c2] : 0.13957     Width : 2.5284e-17
Lifetime [nsec] : 26.033
Charge [e]: 1
Spin : 0/2
Parity : -1
Charge conjugation : 0
Isospin : (I,Iz): (2/2 , 2/2 )
GParity : -1
Quark contents     (d,u,s,c,b,t) : 0, 1, 0, 0, 0, 0
AntiQuark contents               : 1, 0, 0, 0, 0, 0
Lepton number : 0 Baryon number : 0
Particle type : meson [pi]
G4DecayTable:  pi+
0:  BR:  1  [Phase Space]   :   mu+ nu_mu
```

Geant4 particles and hadrons (3/3)

Hadrons are particles interacting with nuclei via the strong interaction
They play a crucial role in several physics simulation domains as collider physics (*e.g. hadronic jets*), nuclear physics (*e.g. neutron shielding*) and atmospheric physics (*e.g. cosmic-ray showers*)

- In `Geant4` with *hadronic physics* we refer (with few exceptions) to the processes: **hadron + nucleus → X**

Bash - Example: inspect proton hadronic processes

```
/particle/select proton # UI command
/particle/process/dump  # UI command

G4ProcessManager: particle[proton]
[0]=== process[Transportation :Transportation] Active
[1]=== process[msc :Electromagnetic] Active
[2]=== process[hIoni :Electromagnetic] Active
[3]=== process[hBrems :Electromagnetic] Active
[4]=== process[hPairProd :Electromagnetic] Active
[5]=== process[CoulombScat :Electromagnetic] Active
[6]=== process[hadElastic :Hadronic] Active
[7]=== process[protonInelastic :Hadronic] Active
```

Hadronic interactions (1/3)

The most accurate theoretical description of hadronic interactions comes from QCD

Good example, equation of cross section calculation for proton-proton interaction at the LHC:

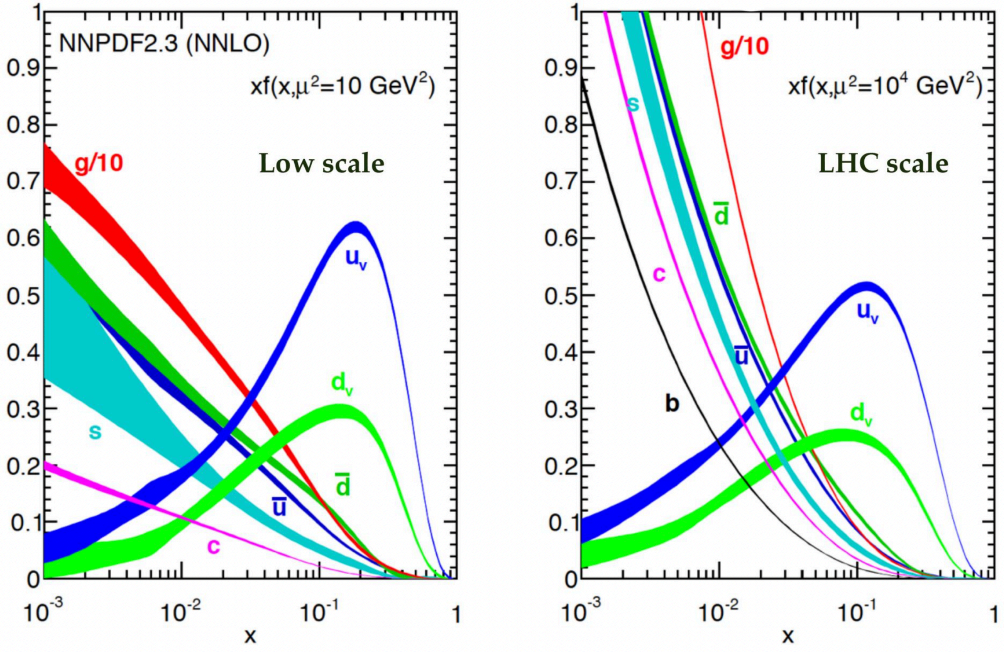
$\hat{\sigma}_{ab\to X}(x_1 x_2 s, Q^2)$:
cross section for partons process

$$\sigma_X(s, Q^2) = \sum_{a,b} \int_{x_{min}}^{1} dx_1 dx_2 f_a(x_1, Q^2) f_b(x_2, Q^2) \hat{\sigma}_{ab\to X}(x_1 x_2 s, Q^2)$$

$f_a(x, Q^2)$:
number of a-type partons with momentum fraction x

$f_b(x, Q^2)$:
number of b-type partons with momentum fraction x

Example of Parton Distribution Functions at different transfer momentum



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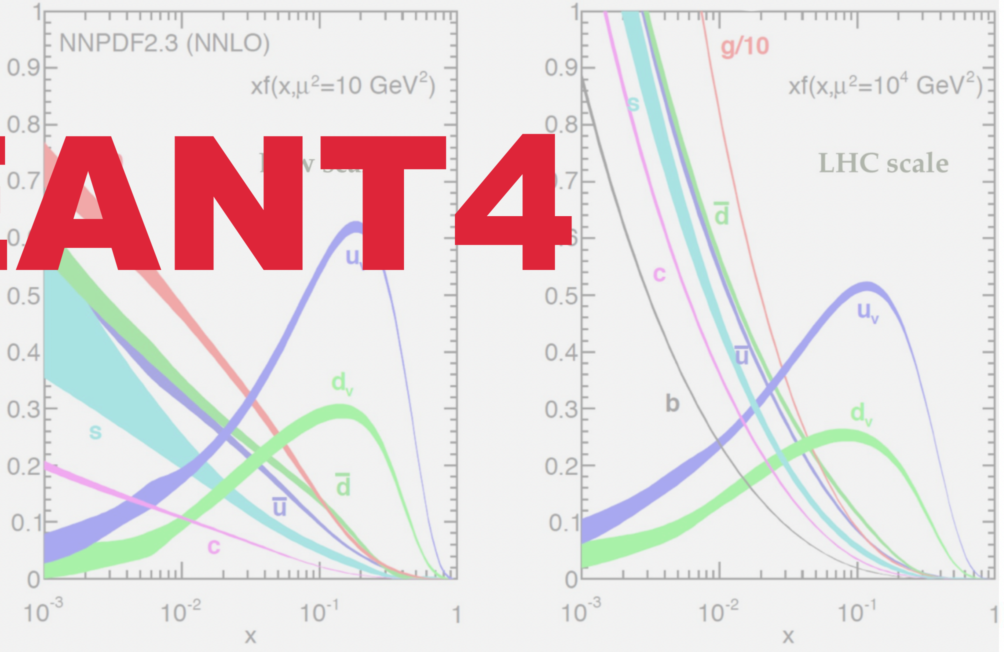
NOT IN GEANT4

$f_a(x, Q^2)$: number of a-type parton with momentum fraction x

$f_b(x, Q^2)$: number of b-type parton with momentum fraction x

Example of Parton Distribution Functions at different transfer momentum



Hadronic interactions (3/3)

- QCD is only applicable through perturbative calculations of hard-scattering with high transverse momentum
- Equivalently, nucleons parton distribution functions are only valid for high transverse momentum interactions
- Therefore, `Geant4` relies on approximate hadronic models with limited ranges of applicability depending on the particle type, energy and target material

Hadronic cross sections

First approximation, `Geant4` separates hadronic cross section datasets from hadronic final states models

- Hadronic cross section datasets refer to the *total cross section* for a hadronic-nucleus interaction (`G4HadronicProcess::GetElementCrossSection()`)
    - For each combination of particle, energy and target-material ≥ 1 cross sections must be specified in a physics list (in case more than one is available the last one set is used)
- In `Geant4` there are only 2 types of hadronic cross sections (neutrons are an exception):
    - The elastic cross section describing the process for which the projectile and the target nucleus survive and no additional particles are created
    - The inelastic cross section describing the process for which any other final state is created

NOTE: there are no *differential cross sections* for hadronic interactions. To get the differential cross section for secondaries in a given phase space one has to multiply the total cross section by the fraction of events corresponding to that final state

RunAction.cc - Example: get hadronic inelastic cross section (without simulation)

```
#include "G4UnitsTable.hh"
#include "G4Element.hh"
#include "G4Material.hh"
#include "G4NistManager.hh"
#include "G4HadronicProcessStore.hh"

void RunAction::EndOfRunAction(const G4Run * /*run*/) {
  if (isMaster) {
    G4HadronicProcessStore *hadstore = G4HadronicProcessStore::Instance();
    G4Material *matPb = G4NistManager::Instance()->FindOrBuildMaterial("G4_Pb");
    G4Element *element = matPb->GetElement(0);
    G4double InelXS1 = hadstore->GetInelasticCrossSectionPerAtom(
        G4Proton::Proton(), 10. * GeV, element, matPb);
    G4cout << "InelXS 10 GeV p on Pb " << G4BestUnit(InelXS1,"Surface") << G4endl;
  }
}
```

Bash - execution

```
InelXS 10 GeV p on Pb 1.83 barn
```

Available on GitHub

Good example of code *modularity*: `Geant4` is a toolkit, not a software

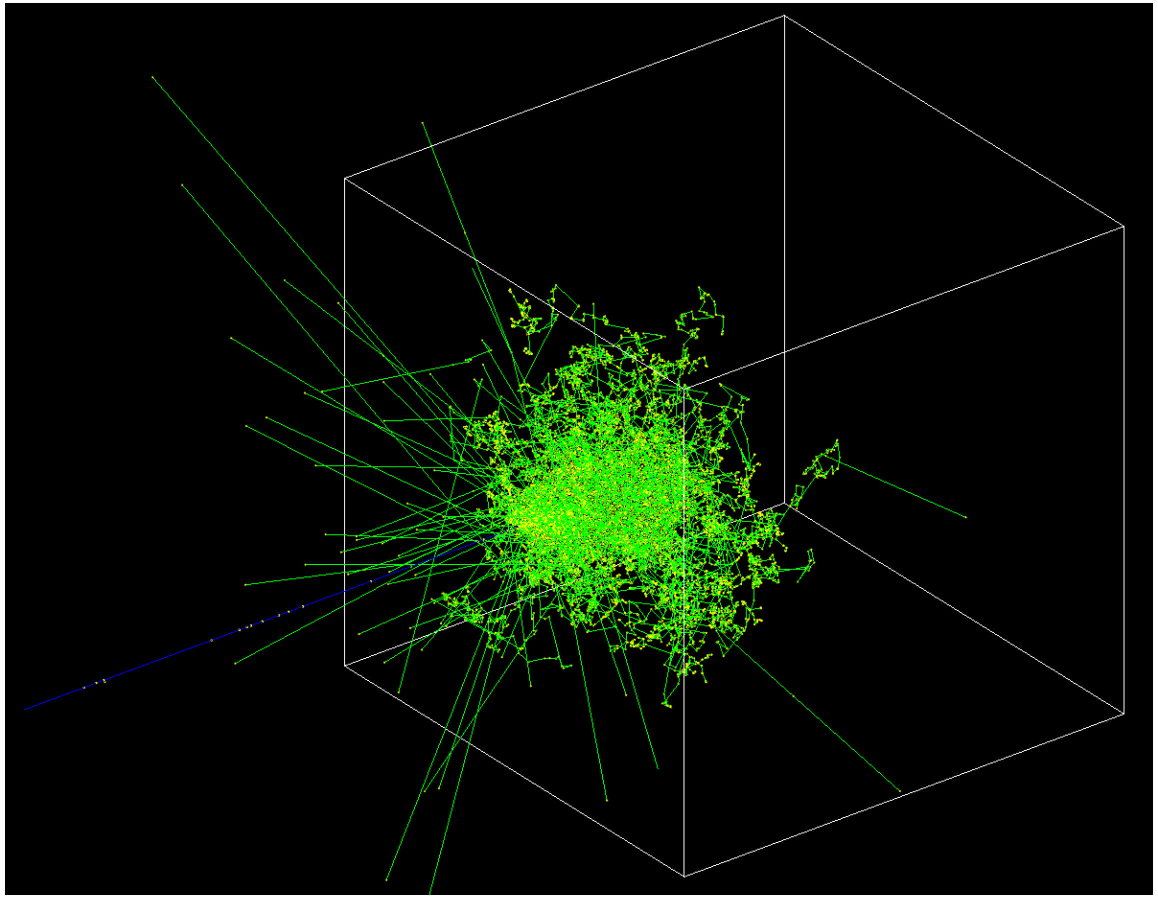
Hadronic final states models

First approximation, `Geant4` separates hadronic cross section datasets from hadronic final states models

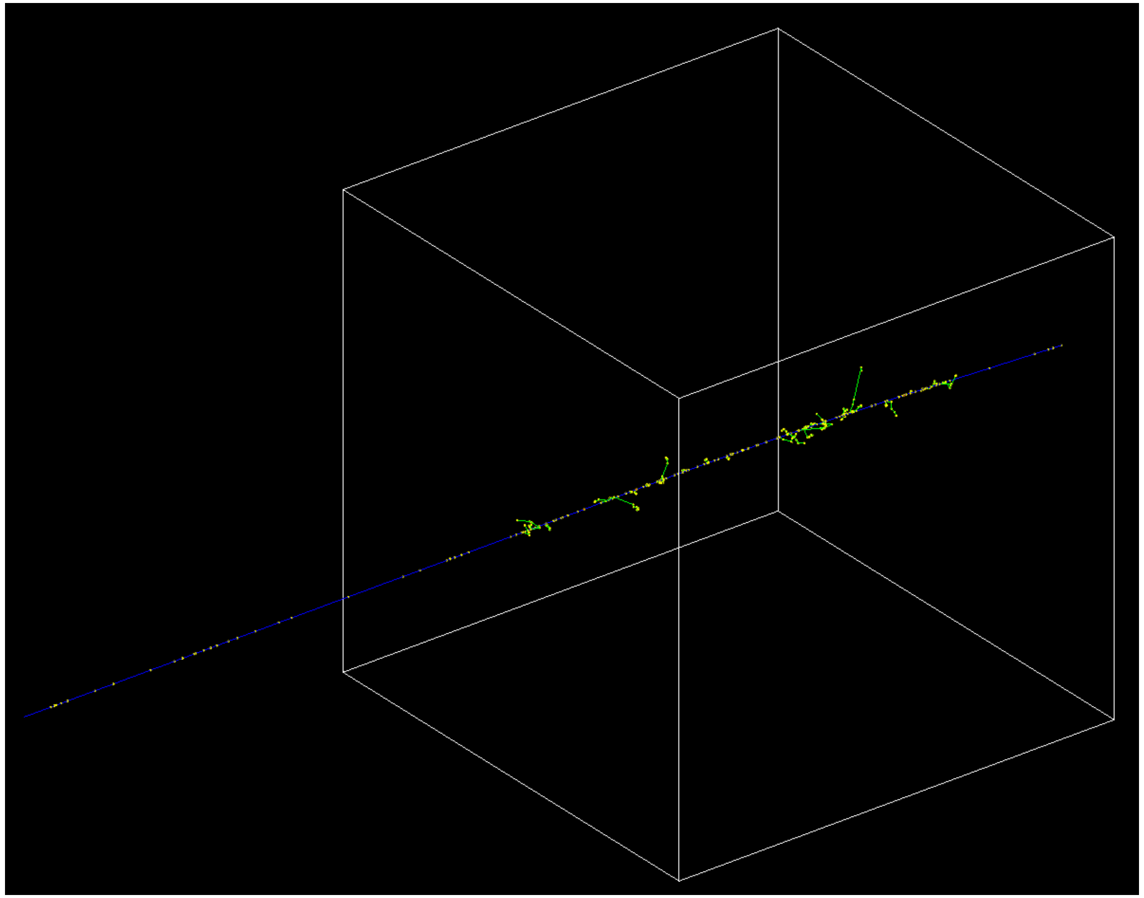
- Hadronic final states models describe the properties of the *secondaries* from the hadronic interaction (`G4HadronicProcess::GetHadronicInteraction()`)
    - For each combination of particle, energy and target-material, 1 or 2 final states model(s) must be specified in a physics list
      (in case two are present in the same energy range, one is selected according to a linear probability interpolation)
- The hadron elastic process competes with the hadron inelastic one
  (as with any other process, e.g. ionization, bremsstrahlung, transportation, …)
- When an inelastic hadronic process occurs, i.e. the process is selected against the others,
  a final state (distribution of secondaries) is produced by one of the hadronic models working in the current energy range

Example: inactivate hadron inelastic (p on Fe)

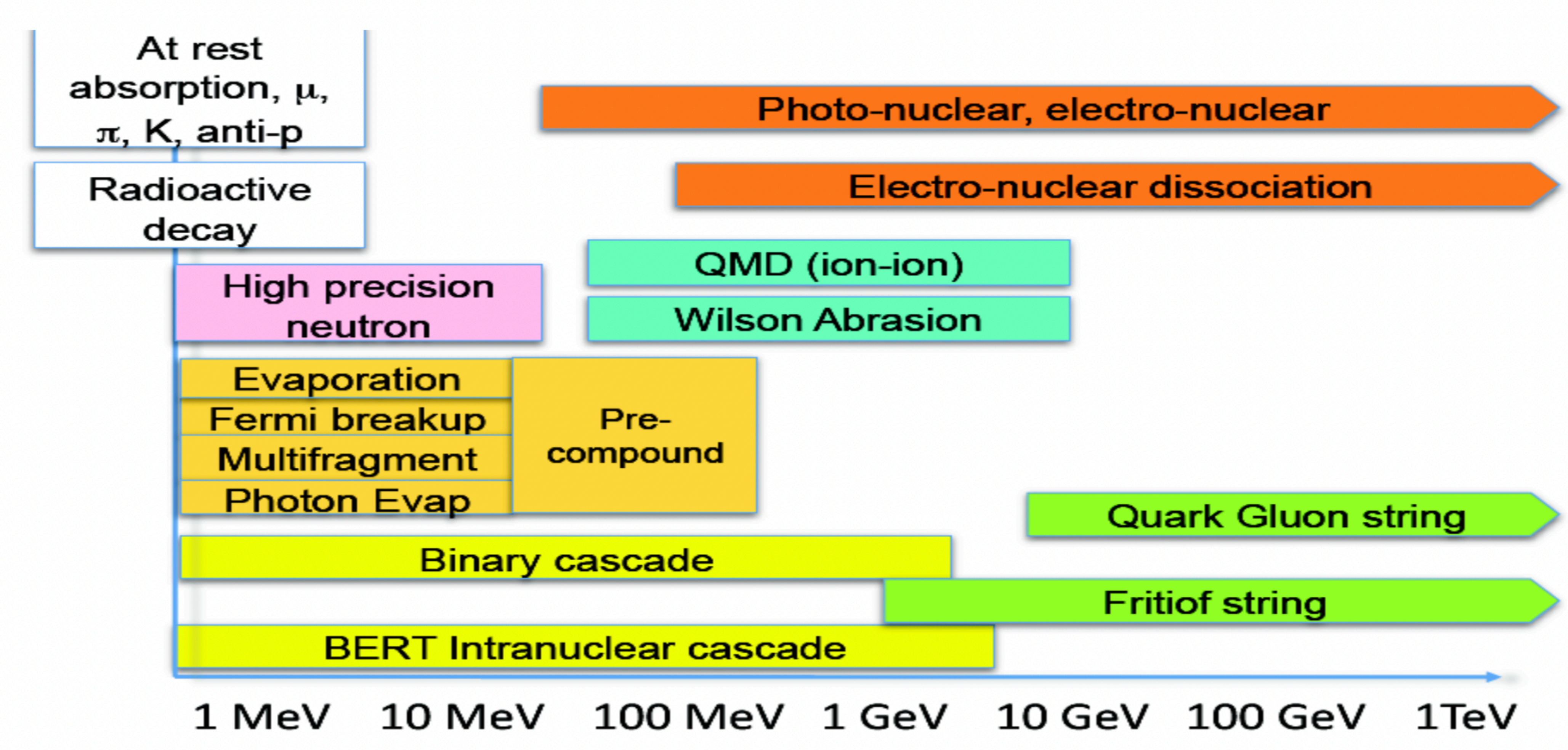
```
/gun/particle proton
/gun/energy 10 GeV
```



```
/gun/particle proton
/gun/energy 10 GeV
/process/inactivate protonInelastic proton
```

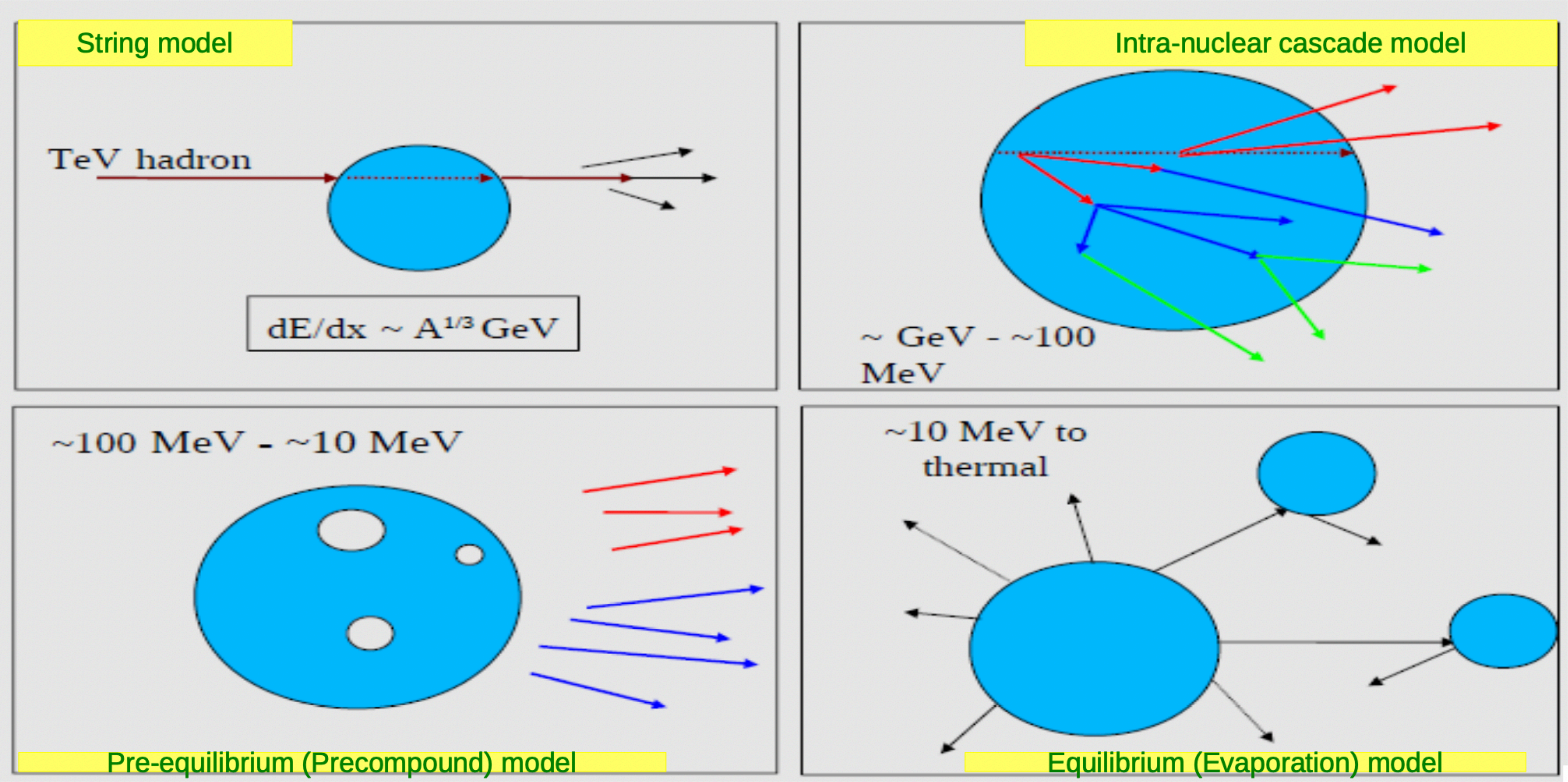


Partial hadronic final states models inventory

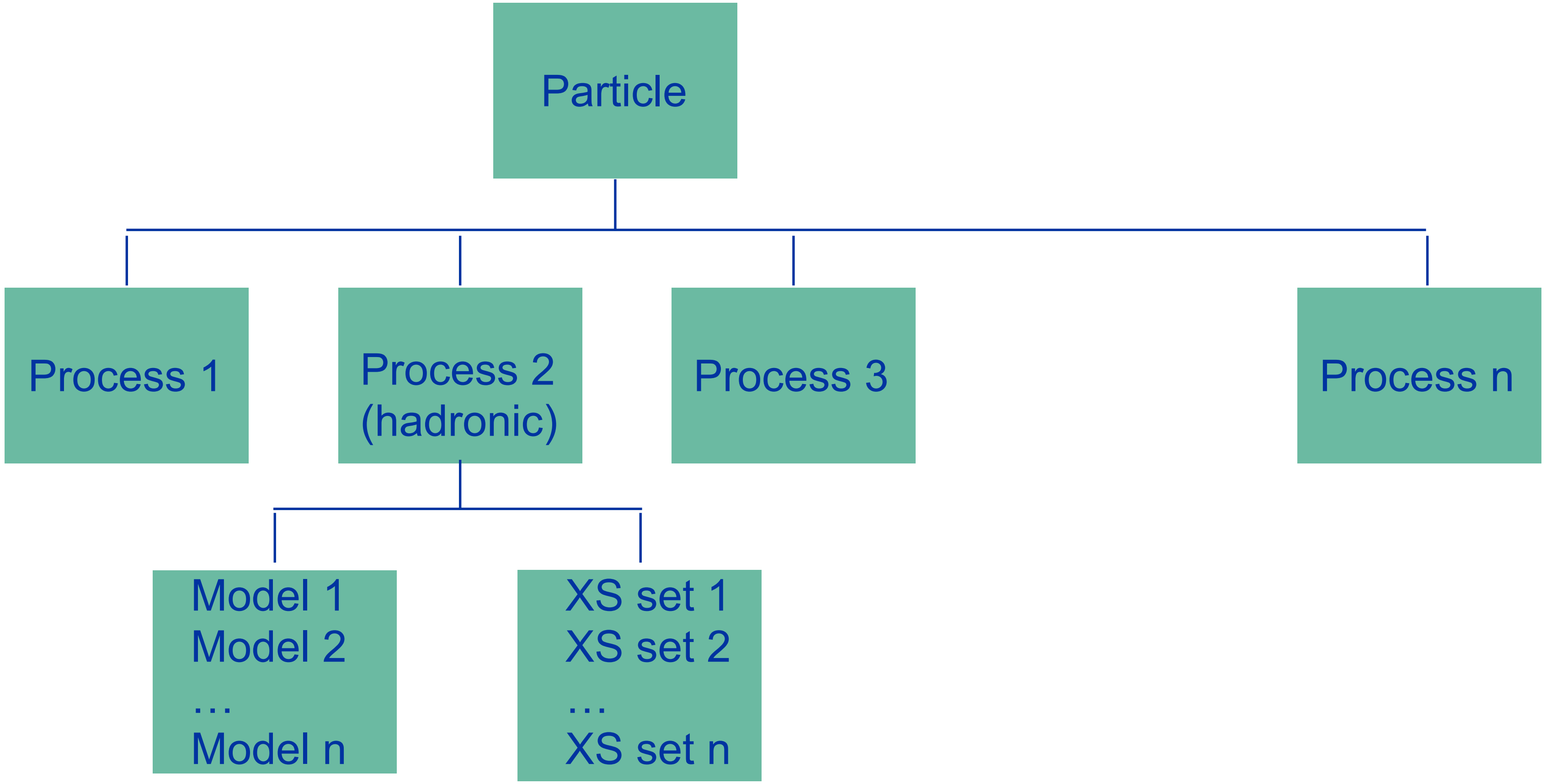


NOTE: Physics list names are derived from the hadronic models used (see lesson *how to choose a physics list*)

Sketch of hadronic final states models

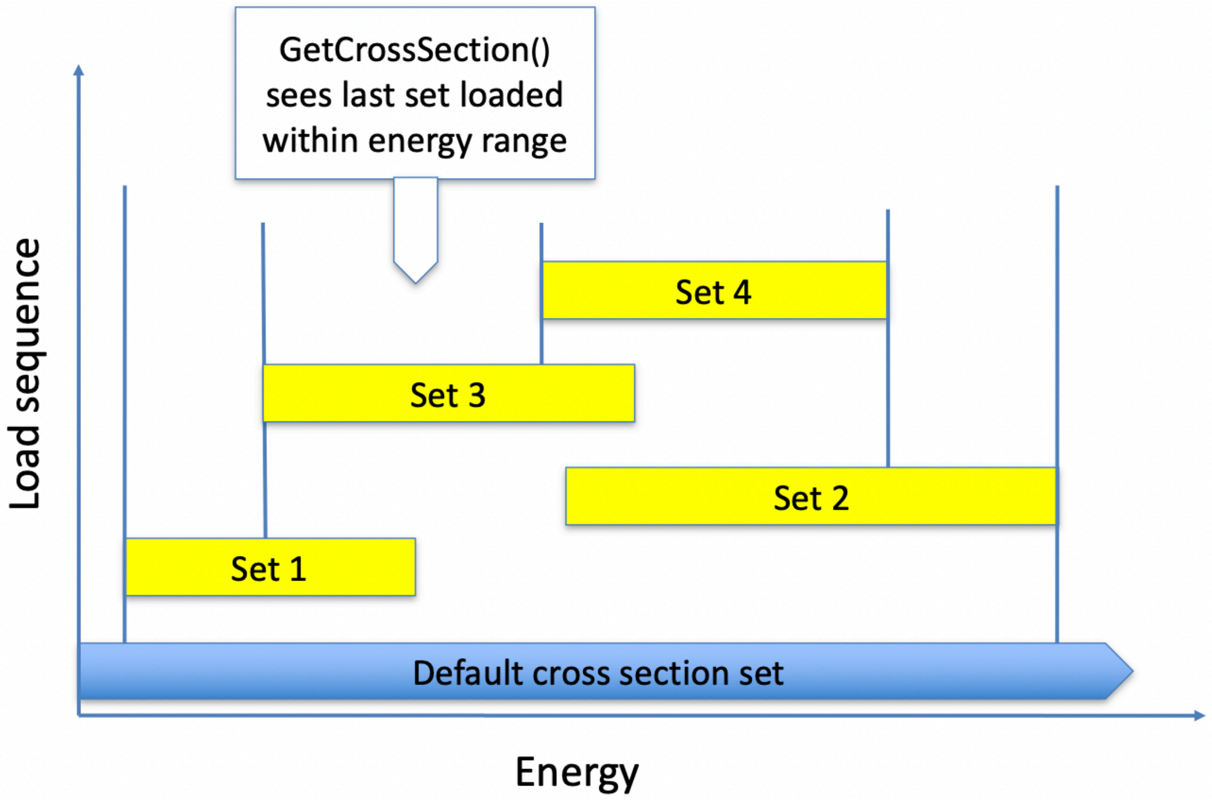


Recap: hadronic physics framework (1/2)

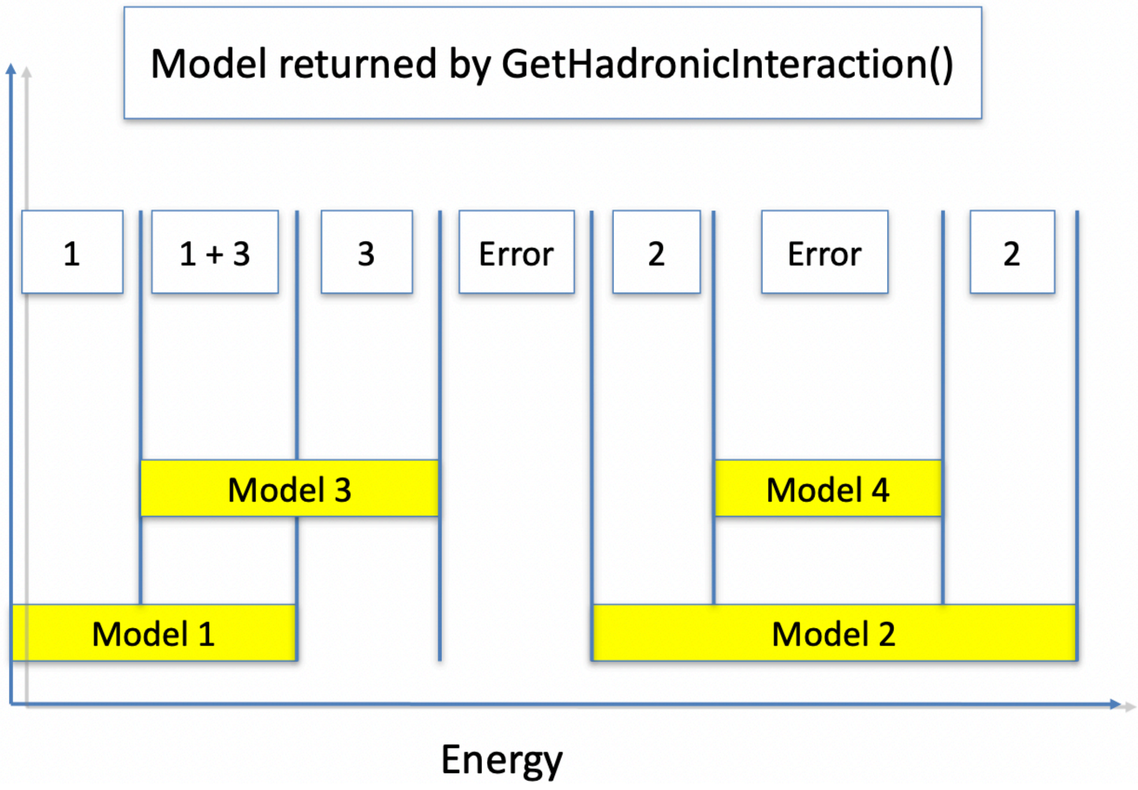


Recap: hadronic physics framework (2/2)

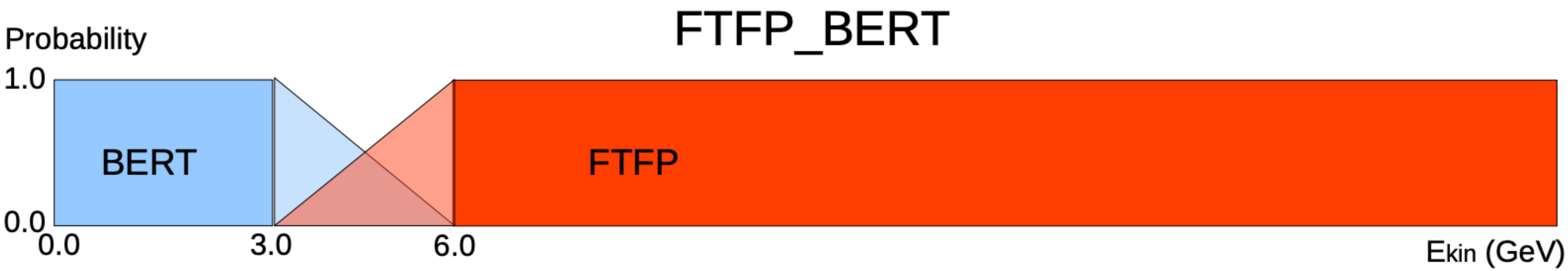
Cross section management

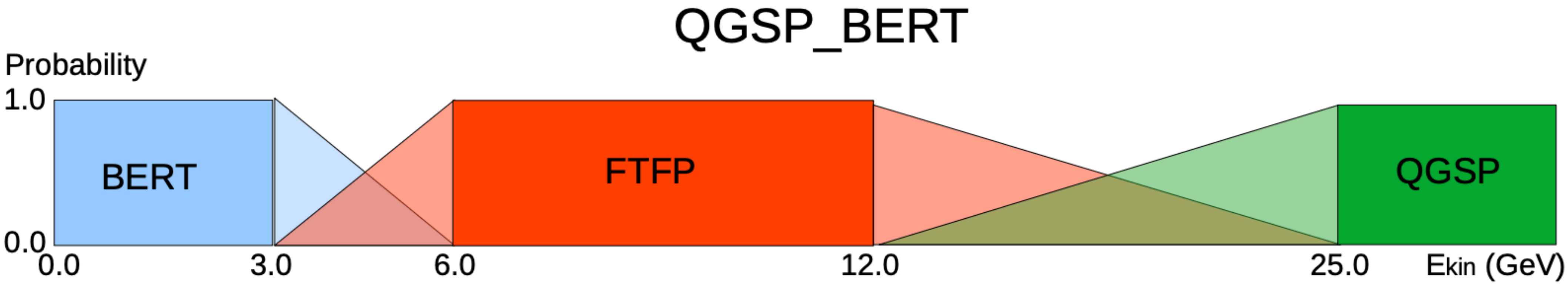


Final state model management



Example of final-state models transition





Nuclear capture at rest

- In `Geant4` charged particles can come at rest, i.e. kinetic energy = 0, due to energy losses by ionization (the simulation is realistic only above few keV)

- Negatively charged hadrons and muons *at-rest* might undergo an additional process, called (nuclear) *capture at rest,* for which they are absorbed by a nucleus
  - It should not be confused with hadron inelastic process happening *in-flight*

Bash - Example: inspect π^- processes

```
/particle/select pi-    # UI command
/particle/process/dump  # UI command

G4ProcessManager: particle[pi-]
[0]=== process[Transportation :Transportation] Active
[1]=== process[msc :Electromagnetic] Active
[2]=== process[hIoni :Electromagnetic] Active
[3]=== process[hBrems :Electromagnetic] Active
[4]=== process[hPairProd :Electromagnetic] Active
[5]=== process[CoulombScat :Electromagnetic] Active
[6]=== process[Decay :Decay] Active
[7]=== process[hadElastic :Hadronic] Active
[8]=== process[pi-Inelastic :Hadronic] Active
[9]=== process[hBertiniCaptureAtRest :Hadronic] Active
```

- The probability for at-rest processes to occur is given by its lifetime (instead of the cross section)

- The nuclear capture at rest processes competes with the decay process

Example: inspect steps of a 10 MeV proton in iron and compare with π^- (next slide)

```
/gun/particle proton
/gun/energy 10 MeV
/tracking/verbose 2
/run/beamOn 1

*****************************************************************************************

* G4Track Information:   Particle = proton,   Track ID = 1,   Parent ID = 0

*****************************************************************************************

Step#     X(mm)     Y(mm)     Z(mm) KinE(MeV)  dE(MeV) StepLeng TrackLeng  NextVolume ProcName

    0         0         0         0        10        0        0         0       World initStep

    1         0         0     0.182      4.92     5.08    0.182     0.182       World hIoni

    2    0.0104   -0.0106     0.261    0.0609     4.86   0.0805     0.263       World hIoni

    3    0.0104   -0.0108     0.261    0.0125   0.0484 0.000532     0.263       World hIoni

    4    0.0104   -0.0108     0.261         0   0.0125 0.000237     0.263       World hIoni

Run terminated.
```

Example: inspect steps of a 10 MeV proton (previous slide) in iron and compare with π^-

```
/gun/particle pi-
/gun/energy 10 MeV
/tracking/verbose 2
/run/beamOn 1
************************************************************************************************

* G4Track Information:   Particle = pi-,   Track ID = 1,   Parent ID = 0

************************************************************************************************
Step#    X(mm)    Y(mm)    Z(mm) KinE(MeV)  dE(MeV) StepLeng TrackLeng  NextVolume ProcName

    0        0        0        0        10        0        0         0       World initStep

    1        0        0    0.364      7.87     2.13    0.366     0.366       World hIoni

    2   0.0197  0.00231    0.652      6.03     1.84    0.291     0.657       World hIoni

    3   0.0132  -0.0289     0.88      3.92     2.11    0.232     0.889       World hIoni

    4  -0.0101  -0.0447     1.04      1.74     2.18    0.164      1.05       World hIoni

    5  -0.0275  -0.0478     1.09    0.0177     1.72   0.0529      1.11       World hIoni

    6  -0.0275  -0.0478     1.09         0   0.0177 0.000351      1.11       World hIoni

    7  -0.0275  -0.0478     1.09         0        0        0      1.11       World hBertiniCaptureAtRest

    :----- List of 2ndaries - #SpawnInStep= 23(Rest=23,Along= 0,Post= 0), #SpawnTotal= 23 ---------------

    :   -0.0275   -0.0478   1.09   0.00968                  e-
```

A special case: neutrons (1/2)

- Neutrons are crucial particles for every simulation involving hadronic physics (*e.g. hadronic calorimeters, nuclear reactors, shielding, …*)
- Typically several *soft* neutrons are released from nuclei de-excitation after the hadron inelastic process (take care *lepto-nuclear* and *gamma-nuclear* processes release neutrons too)

Bash - Example: inspect neutron processes (FTFP_BERT)

```
/particle/select neutron # UI command
/particle/process/dump   # UI command

G4ProcessManager: particle[neutron]
[0]=== process[Transportation :Transportation] Active
[1]=== process[Decay :Decay] Active
[2]=== process[hadElastic :Hadronic] Active
[3]=== process[neutronInelastic :Hadronic] Active
[4]=== process[nCapture :Hadronic] Active
[5]=== process[nKiller :General] Active
```

- The `nCapture` (radiative capture) process can happen *in-flight* (differently from the radiative capture at rest for negatively charged hadrons), the neutron is absorbed by a nucleus which emits γ (see next slide)

NOTE: In hadronic showers neutrons are typically the third most abundant particles produced, after γ and e^-

Example: inspect steps of a 1 MeV neutron in iron (FTFP_BERT)

```
/gun/particle neutron
/gun/energy 1 MeV
/tracking/verbose 2
/run/beamOn 1
*************************************************************************************************

* G4Track Information:   Particle = neutron,   Track ID = 1,   Parent ID = 0

*************************************************************************************************
Step#    X(mm)    Y(mm)    Z(mm) KinE(MeV)  dE(MeV) StepLeng TrackLeng  NextVolume ProcName

    0        0        0        0         1        0        0         0       World initStep

    1        0        0    0.311     0.962   0.0384    0.311     0.311       World hadElastic

    2    -51.8     34.7    -4.95      0.94   0.0213     62.5      62.9       World hadElastic

    3    -61.4     50.3    -57.5     0.931  0.00918     55.6       118       World hadElastic

...

  170     27.6      242      270         0        0     6.24  3.86e+03       World nCapture

    :----- List of 2ndaries - #SpawnInStep=  2(Rest= 0,Along= 0,Post= 2), #SpawnTotal=  2 -----
    :      27.6       242       270      7.63              gamma
    :      27.6       242       270  0.000503       Fe57[14.413]
    :----------------------------------------------------------------- EndOf2ndaries Info -----
```

A special case: neutrons (2/2)

- As neutrons do not loose energy by ionization, they usually undergo many elastic collisions (eventually down to thermalization) with nuclei before being killed
- In `Geant4` neutrons can be killed by the hadron inelastic process, the decay process, the radiative capture process, the `nKiller` process (default threshold $10\mu s$) or by the Geant4 kernel (*out-of-world killed*)
- Neutron cross sections are wild: they depend, sometimes dramatically and unpredictably, on the neutron energy and the target element/isotope
- The CPU time varies up to one order of magnitude depending on the accuracy of neutron simulation
  - For most high-energy physics and nuclear applications, a simplified and fast description is usually sufficient (`FTFP_BERT, QGSP_BERT, …`)
  - Where more accuracy is needed, `Geant4` offers a more precise, data-driven and isotope-specific treatment for neutrons, for kinetic energies < 20 MeV (see lesson *Hadronic Physics II*)

Datasets

Some hadronic models, for both final states and cross sections, are data-driven, *i.e.* they need as input phenomenological data; similarly, some need as input the result of intensive computation that are performed before the simulation

- The `-DGEANT4_INSTALL_DATA=ON` cmake option sets the datasets automatic download and installation
- The following `envs` are exported by `geant4.sh` and linked to datasets

geant4install/bin/geant4.sh - [Mandatory to every physics list, needed only to specific physics lists]

```
# Resource file paths
# - Datasets

export G4NEUTRONHPDATA="/path-to/geant4-install/share/Geant4/data/G4NDL4.7"
export G4LEDATA="/path-to/geant4-install/share/Geant4/data/G4EMLOW8.2"
export G4LEVELGAMMADATA="/path-to/geant4-install/share/Geant4/data/PhotonEvaporation5.7"
export G4RADIOACTIVEDATA="/path-to/geant4-install/share/Geant4/data/RadioactiveDecay5.6"
export G4PARTICLEXSDATA="/path-to/geant4-install/share/Geant4/data/G4PARTICLEXS4.0"
export G4PIIDATA="/path-to/geant4-install/share/Geant4/data/G4PII1.3"
export G4REALSURFACEDATA="/path-to/geant4-install/share/Geant4/data/RealSurface2.2"
export G4SAIDXSDATA="/path-to/geant4-install/share/Geant4/data/G4SAIDDATA2.0"
export G4ABLADATA="/path-to/geant4-install/share/Geant4/data/G4ABLA3.1"
export G4INCLDATA="/path-to/geant4-install/share/Geant4/data/G4INCL1.0"
export G4ENSDFSTATEDATA="/path-to/geant4-install/share/Geant4/data/G4ENSDFSTATE2.3"
```

Datasets (2/3)

- **G4LEDATA** : low-energy electromagnetic data, mostly derived from Livermore data libraries; used in all EM options
- **G4LEVELGAMMADATA** : photon evaporation data, come from the Evaluated Nuclear Structure Data File (ENSDF); used by Precompound/de-excitation models (and RadioactiveDecay if present)
- **G4SAIDXSDATA** : data evaluated from the SAID database for nucleon and pion cross sections below 3 GeV; used in all physics lists
- **G4PARTICLEXSDATA** : evaluated neutron (as well as proton, deuteron, triton, He3, alpha, gamma, neutrino) cross sections derived from G4NDL by averaging in bin of energies; used in all physics lists
- **G4ENSDFSTATEDATA** : nuclear properties, from Evaluated Nuclear Structure Data File (ENSDF); used in all physics lists

Datasets (3/3)

- **G4REALSURFACEDATA** : data for measured optical surface reflectance look-up tables; used only when optical physics is activated
- **G4NEUTRONHPDATA** : evaluated neutron data of cross sections, angular distributions and final-state information; come largely from the JEFF-3.3 library; used only in _HP physics lists (see *Hadronic Physics II* at this course)
- **G4RADIOACTIVEDATA** : radioactive decay data, come from the ENSDF; used only when radioactive decay is activated (see Hadronic Physics II at this course)
- **G4INCLDATA** : data for the intranuclear cascade model INCLXX
- **G4ABLADATA** : data for the ABLA de-excitation model, which is an alternative de-excitation available for INCLXX

Physics Lists

geant4/source/physics_lists/lists/src/FTFP_BERT.cc

- A *physics list* is a class that constructs all the particles and the associated processes
- One and only one physics list must be present in a `Geant4` simulation
- Using a reference physics list is straightforward, *e.g.* in `main()`

```
G4VModularPhysicsList* PL = new FTFP_BERT();
runManager->SetUserInitialization(PL);
```

- No default physic list (and no default physics) is provided by `Geant4`, the user is responsible to choose or create a Physics List
- A modular physics list implements a granular approach to physics via *constructors*
- Example of hadronic physics constructors

```
FTFP_BERT::FTFP_BERT(G4int ver){
  if(ver > 0) {
    G4cout << "<<< Geant4 Physics List simulation engine: FTFP_BERT"<<G4endl;
    G4cout <<G4endl;
  }
  defaultCutValue = 0.7*CLHEP::mm;
  SetVerboseLevel(ver);

  // EM Physics
  RegisterPhysics( new G4EmStandardPhysics(ver));
  // Synchroton Radiation & GN Physics
  RegisterPhysics( new G4EmExtraPhysics(ver) );
  // Decays
  RegisterPhysics( new G4DecayPhysics(ver) );
  // Hadron Elastic scattering
  RegisterPhysics( new G4HadronElasticPhysics(ver) );
  // Hadron Physics
  RegisterPhysics(  new G4HadronPhysicsFTFP_BERT(ver));
  // Stopping Physics
  RegisterPhysics( new G4StoppingPhysics(ver) );
  // Ion Physics
  RegisterPhysics( new G4IonPhysics(ver));
  // Neutron tracking cut
  RegisterPhysics( new G4NeutronTrackingCut(ver));
}
```

Reference physics lists (1/2)

The **`FTFP_BERT` PL** is the recommended one for high-energy physics simulations. It adopts the models:

- FTF: (Fritiof) hadronic string model for hadron inelastic process > 3 GeV
- P: Precompound model for nucleus de-excitation + evaporation models
- BERT: (Bertini) intra-nuclear cascade model for the hadron inelastic process < 6 GeV
- Plus: neutron radiative capture, nuclear capture at rest for negatively charged hadrons, elastic scattering for hadrons, lepto-nuclear and gamma-nuclear processes and standard em physics (*compton, pair-production, photoelectric effect, …*)
- No high-precision treatment of low-energy neutrons (explained in *hadronic physics II*), no radioactive decays (explained in *hadronic physics II*), no optical photons

NOTE: ATLAS, CMS, LHCb and ALICE use mild-variants (tuning) of this physics list
(e.g. the `FTFP_BERT_ATL` physics list used by the ATLAS Experiment)

Reference physics lists (2/2)

Other relevant reference physics lists for hadronic physics are

- The **`FTFP_BERT_HP`** PL: same as `FTFP_BERT` but using the high-precision neutron treatment for low energy neutrons (< 20 MeV)
- The **`QGSP_BERT`** PL: same as `FTFP_BERT` but uses the QGS (Quark Gluon String) model for the hadron inelastic process with an energy transition range with FTF of [12,25] GeV

  NOTE: Below 12 GeV `QGSP_BERT` and `FTFP_BERT` are identical
- The **`FTFP_INCLXX`** PL: uses the INCLXX model for the inelastic process of protons, neutrons and charged pions. The INCLXX and the FTF models intersect in the energy range [15,20] GeV
- The **`QGSP_BIC`** PL: uses both FTF and QGS models for high-energy interactions and both BERT and BIC (Binary Cascade) models for low-energy interactions, as
  - Protons, neutrons: BIC < 6 GeV, FTFP $[3{,}25]$ GeV and QGSP > 12 GeV
  - Pions and kaons: BERT < 6 GeV, FTFP $[3{,}25]$ GeV and QGSP > 12 GeV

Recap of *Hadronic Physics I*

- `Geant4` implements two main hadronic processes, the *hadron elastic* and the *hadron inelastic* processes
- For negatively charged hadrons the *nuclear capture at rest* process is included as well
- For neutrons the *radiative capture* process is also included
- In `Geant4` cross section datasets and final states models are clearly separated (no differential cross sections available)
- Several final state models are available and applicable to restricted combinations of energies, particles and materials
- Reference physics lists construct hadronic physics by picking up both hadronic final states models and cross section datasets
  - Typically reference physics lists are named after the hadronic final state models adopted

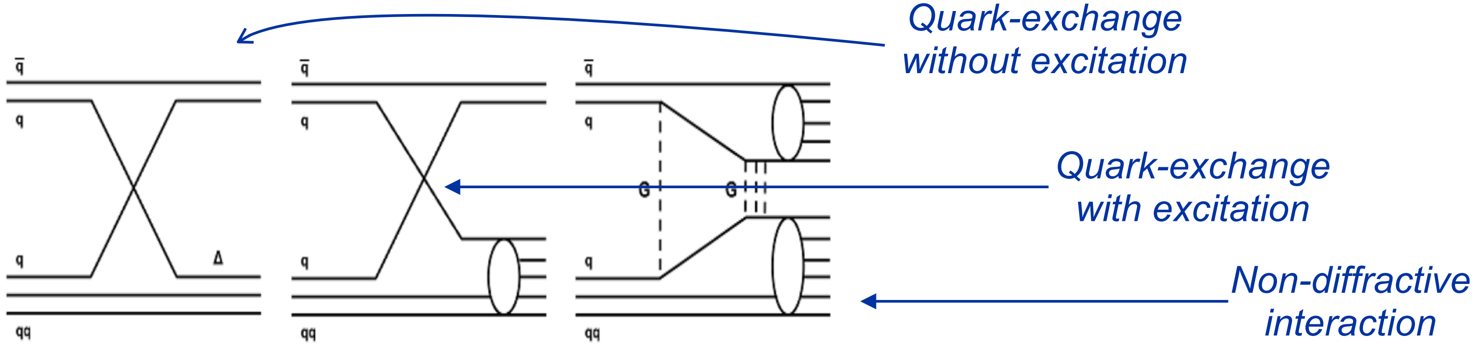
Backup material

The FTF String model algorithm (1/2)

- Parton string models in `Geant4` describe the inelastic interaction of on nucleon/nuclear-target with nucleon, π and k projectiles. The interaction is described in terms of constituent quarks
- The target nucleus (A nucleons, Z protons) is initialized by selecting r_i coordinates and p_i momenta with $i = 1,\ldots,A$
  - The energy per nucleon is computed as $e = E/A = m_N + B(A,Z)/A$ where m_N is the nucleon mass and $B(A,Z)$ is the nucleon binding energy.
    - The effective mass of each nucleon is $m_{eff} = \sqrt{(E/A)^2 - p_i^2}$
- The *impact parameter* of the projectile over the target is randomly chosen
- The interaction point z_i is sampled according to the total projectile-nucleon cross-section
  - The closest nucleon is chosen as the interacting one
  - A choice between elastic and inelastic interaction is implemented

The FTF String model algorithm (2/2)

- In case an inelastic collision is chosen a multi-particle process is simulated:
  - The FTF model assumes that all hadron-nucleon interactions are binary reactions $h_1 + h_2 \rightarrow h_1' + h_2'$
- The hadron nucleon interaction can be a quark exchange with excitation, quark exchange without excitation, or a diffractive/non-diffractive interaction
  - Examples for $\pi^- p$ interaction:



- For each produced particle new interaction points are sampled
- Excited hadrons, referred to as *strings*, is decayed into stable particles by the LUND fragmentation model
- The excitation energy of the residual nucleus is calculated and the nucleus is de-excited with the Precompound and evaporation models